ABHAY KUMAR

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EDUCATION

National Institute of Technology Raipur 2023 - 2027 (Expected)
B.Tech in Computer Science and Engineering **CGPA: 8/10**

TECHNICAL SKILLS

Languages: C++, Python, TypeScript, JavaScript, SQL
Backend: Node.js, Express.js, Fastify, FastAPI, REST APIs, Socket.IO, JWT, asynchronous processing
Databases & Queues: PostgreSQL, MongoDB, Redis, BullMQ, MikroORM
Testing & DevOps: Vitest, Postman, Zod, Git, Docker, GitHub Actions, CI/CD, Linux
CS Fundamentals: Data structures, algorithms, OOP, operating systems, DBMS, computer networks

EXPERIENCE & OPEN SOURCE

Endora Commerce | Open-Source Contributor Oct 2026
- Merged a TypeScript backend fix for configurable retention in single-asset and recursive-folder deletion; corrected global settings resolution while preserving existing deadlines and idempotent repeat deletes. PR #328
- Added 31 Vitest regression cases, including HTTP/PostgreSQL tests for retention boundaries, defaults, and nested folders; validated 105 targeted tests locally, with all 9 recorded GitHub checks passing.
- Previously fixed newsletter API serialization with MikroORM and Zod, preserving nullable responses; added 2 PostgreSQL integration tests for entity reloads and schema validation. Merged PR #266

AshwaQuant Pvt. Ltd. | Software Developer Intern May 2025 - Jul 2025
- Built a full-stack course platform for a trading-education startup, supporting enrollment and onboarding.
- Owned Razorpay payment initiation and verification, integrating successful payments with course onboarding.
- Developed and debugged Node.js/Express REST APIs; tested endpoints with Postman and managed source changes with Git.

PROJECTS

EventForge - Event Processing and Job Queues GitHub | Live
TypeScript, Fastify, MongoDB, Redis, BullMQ, React, Docker Compose
- Built multi-tenant REST ingestion APIs and asynchronous workers with idempotency, priority queues, exponential retries, dead-letter recovery, and tenant isolation.
- Built a dashboard for job states, failures, and latency; verified reliability with 8 HTTP correctness probes.
- Benchmarked a reference handler across 3 local Docker runs totaling 1,500 events: 404-712 accepted requests/s and 31-57 ms p95 ingestion latency; all measured events completed.

Chatty AI - AI-Assisted Real-Time Chat GitHub | Live
React, TypeScript, Node.js, Express, Socket.IO, MongoDB, Gemini, Cloudinary
- Built real-time messaging, image sharing, typing indicators, and online presence; persisted messages in MongoDB before Socket.IO delivery.
- Implemented JWT cookie/token authentication, REST validation, and Cloudinary uploads; used Zustand for client state and supported a customizable 9-theme interface.
- Integrated Gemini smart replies, typing autocomplete with 600 ms debouncing, and conversation analysis; added fallbacks for AI service failures.

Pharma - Multi-Agent Research Assistant GitHub | Live
Python, FastAPI, LangGraph, LangChain, RAG, React, Docker
- Built an AI pharmaceutical research assistant for the EY Hackathon, automating drug-disease literature discovery, analysis, and structured report generation.
- Orchestrated research planning, paper retrieval, ranking, and summarization with LangGraph and LangChain; implemented retry/backoff and logging for workflow troubleshooting.
- Integrated RAG with embeddings, semantic search, and vector database retrieval; exposed FastAPI endpoints and a React frontend for exploring research insights.

ACHIEVEMENTS

- **Competitive Programming:** Codeforces Expert (1638), LeetCode Knight (1973), CodeChef 4-Star (1750).
- **Problem Solving:** Solved 1,800+ DSA/CP problems; LeetCode ranks 804 (WC 495) and 311 (BWC 192).
- **Amazon ML Challenge:** AIR 545 (2026, business entity resolution); AIR 818 (2025, multimodal product-price regression).
- **Economic Times Gen AI Hackathon 2025:** Winning team - NewsNavigator.
- **Academic:** GATE Computer Science - AIR 9221; KVPY Scholar 2022 - AIR 2000.
- **Certifications:** Oracle Cloud Infrastructure AI Foundations Associate; AI Skills Passport (EY & Microsoft).